Helmi Fakhriandi Rachman

Software Engineer

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SUMMARY

Software Engineer with 6+ years of backend experience in payments. Since 2020, vendor developer for Artajasa, operator of the ATM Bersama network in Indonesia, building ISO 8583 and QRIS systems in Java and Spring Boot. Designed and built a QRIS certification platform as the main engineer, and now build Oracle Fusion ERP integration in Node.js for a telecom client, with 669 commits shipped to production in 11 weeks. Strong in TDD, SQL tuning and AI-assisted development, with about 3,000 automated tests written across two projects.

SKILLS

Payments: ISO 8583, QRIS, Payment Switching, Payment Gateway, TCP/IP Socket, jPOS, Host Simulator, UAT and Certification Testing

Programming Languages: Java 17, JavaScript, SQL, PHP

Frameworks: Spring Boot, Spring Security, Spring Data JPA, Thymeleaf, Node.js, Express.js, Knex.js, React, Ant Design, Yii 1.1, Laravel

APIs and Integration: Microservices, RESTful API, JWT, HMAC Signing, SOAP, Oracle Fusion ERP Integration

Databases: PostgreSQL, MySQL, Oracle, Redis (basic), SQL Query Tuning

Testing and Tools: JUnit, Jest, Test-Driven Development (TDD), Postman, Git, Claude Code (agentic coding), Graphify, Obsidian

Security: OWASP Top 10, Penetration Test Remediation

Currently Learning: Go, GCP, Docker, CI/CD

Languages: Indonesian (Native), English (Professional)

EXPERIENCE

Software Engineer - Vendor Developer for Artajasa (ATM Bersama Network) | Pusat Ilmu Komputer Universitas Indonesia (Pusilkom UI)

Jakarta, Indonesia | Mar 2020 - Present

- As the main engineer, designed and built a UAT automation and certification platform for ATM Bersama QRIS in Java 17, Spring Boot, Thymeleaf, JPA, PostgreSQL and jPOS (311 of 371 commits; about 39,000 lines in the codebase). Its sandbox host simulator runs an ISO 8583 TCP server and a JWT and HMAC-secured REST API, converts JSON to ISO 8583, and is covered by 766 JUnit tests. The tool replaces manual tests and writes the UAT report automatically.
- Handed the platform to two developers in Aug 2026 and now mentor them and review their code.
- Since Aug 2026, building Oracle Fusion ERP integration for a telecom client's project management system in Node.js, Express.js, Knex, PostgreSQL and React. Shipped 669 commits to production in 11 weeks (455 back end, 214 front end), all test-first with Claude Code: 530 test files with about 2,650 test cases.
- Ran the go-live data migration: built a parallel load runner and loaded about 27,000 ERP files (about 2 million rows for 12,981 purchase orders) in 4 days, with a backup and tested rollback for every step. Made a 20,580-row Excel export 60x faster (16.5 minutes to 15.5 seconds, memory 1.6 GB to 630 MB) and fixed the query behind 49% of production database time (up to 1 hour down to under 4 seconds in staging tests).
- 2025 - 2026: main developer (192 of 322 commits) of a bill-hosting portal for billers in Spring Boot, Thymeleaf, Spring Security, Oracle and PostgreSQL. Built bill upload and validation, payment monitoring, settlement and reconciliation reports, QR code images for payment URLs and broadcast notifications. Fixed about 80 security code review findings (CSP, CSRF, OWASP output encoding, session expiry, Tomcat upgrade).
- 2022 - 2025: main developer (72 of 102 commits) of an e-money back office in Laravel and PostgreSQL, with maker-checker approvals, fraud and blacklist controls and account lockout. Also built settlement and reconciliation features for a disbursement system in Yii 1.1, Oracle and MySQL, with JasperReports and a Java reconciliation scheduler.
- 2020 - 2025: built ISO 8583 switching and payment gateway proxy features in Java over TCP sockets in a microservice setup (separate back-end, worker, front-end, proxy, payment and disbursement APIs). Also built a dashboard for transactions worth billions of rupiah.
- Debugged and tuned slow reports to cut report and page load times from 15-30 minutes to under 5 minutes on up to one million rows of daily data. Fixed 10 to 50 penetration test findings per project, including all high and critical issues.

Back-end Engineer Intern | Jojonomic

Jakarta, Indonesia | Jul 2017 - Oct 2017

- Maintained core APIs and integrated third-party APIs, including SOAP services, as part of the integration team.

EDUCATION

Bachelor of Computer Science (S.Kom) | Universitas Indonesia | 2012 - 2018

Faculty of Computer Science | Activities: Teaching Assistant